

⇒ RECAP

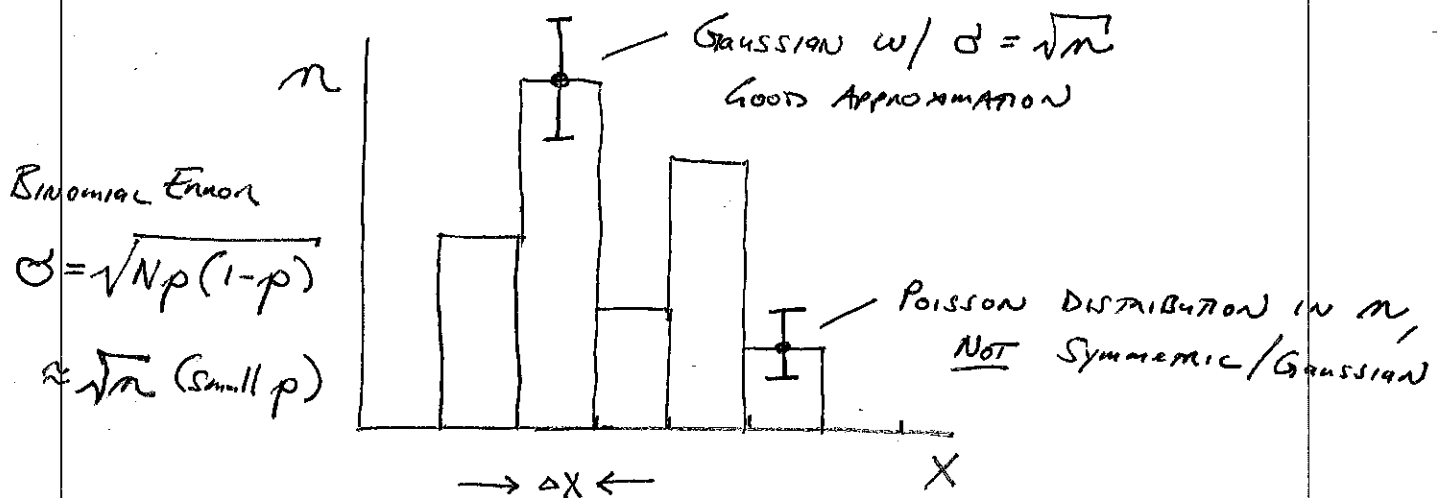
CLASSICAL INFERENCE
(from data)

- ESTIMATE PARAMETERS
- QUANTIFY CONFIDENCE
- CONSISTENCY OF DATA/MODEL

TECHNIQUES

- METHOD OF MOMENTS
(MEAN, STD DEV, SKEWNESS...)
- χ^2 MINIMIZATION / LEAST SQUARES
"FITTING" EPDF, FOR DATA
WITH GAUSSIAN UNCERTAINTIES

HISTOGRAMMED DATA



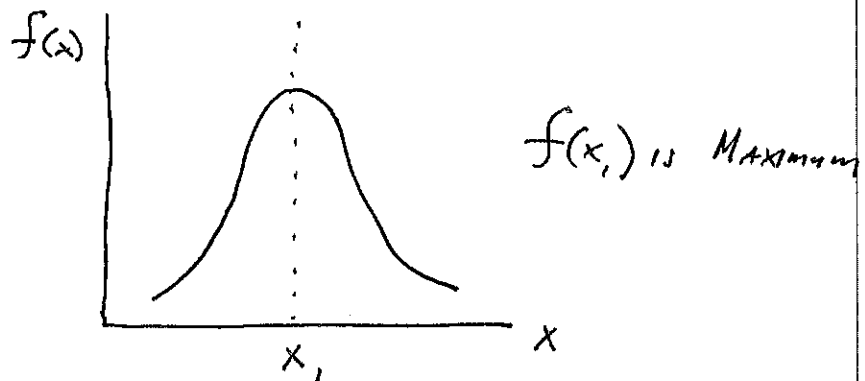
↳ We need a method for fitting
THAT Does Not Assume Gaussian Error. ✓

⇒ Maximum Likelihood Estimation (MLE)

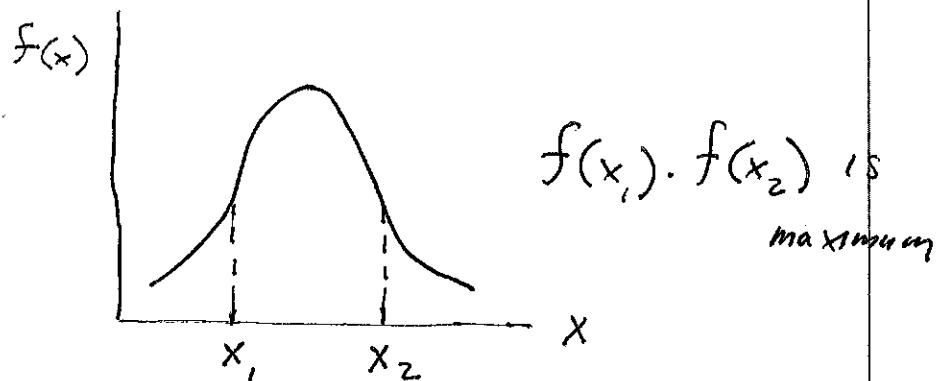
(Gauss, Laplace, Fisher...)

Qualitative Intro:

* Suppose we have one data point drawn from a Gaussian RV. Best guess for mean of that RV is x_1



* What if I have two points $x_1 \neq x_2$



(Suggestion: write a short python script to convince yourself this is true)

* Continuing: Best Guess For N measurements?

$$L = \prod_{i=1}^N f(x_i | M(\theta))$$

Diagram illustrating the components of the Likelihood function L :

- L is labeled "LIKELIHOOD" with an arrow pointing to it.
- The product symbol $\prod$ is labeled "PRODUCT" with an arrow pointing to it.
- The index $i=1$ is labeled "MODEL" with an arrow pointing to it.
- The function $f(x_i | M(\theta))$ is labeled "Parameters" with an arrow pointing to it.

↳ THE BEST VALUE OF θ FOR N MEASUREMENTS WILL BE THAT WHICH MAXIMIZES L

- MLE Method - FORMULATE LIKELIHOOD FOR MODEL M
- FIND PARAMETERS θ THAT MAXIMIZE L
 - DERIVE CONFIDENCE REGION FOR θ
 - PERFORM HYPOTHESIS TESTS

EXAMPLE

Suppose I measure the length of a rod
N times. Suppose measurement has Gaussian,
"HOMOSCEDASTIC" UNCERTAINTIES
ALL THE SAME!

$$L = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i - \mu)^2 / 2\sigma^2}$$

μ IS ... THE TRUE LENGTH OF THE ROD
(WHAT WE HOPE TO ESTIMATE)

→ "GROUND TRUTH"

→ "TRUTH WITH A CAPITAL T"

IN PRACTICE, L IS VERY SMALL & hard to compute.
So we'll take its NATURAL LOGARITHM.

$$\begin{aligned}\ln(L) &= \sum_{i=1}^N \ln \left\{ \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i - \mu)^2 / 2\sigma^2} \right\} \\ &= N \ln \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= [\text{CONSTANT}] - \frac{1}{2} \chi^2\end{aligned}$$

THE MAXIMUM LOG-LIKELIHOOD $\leftrightarrow$ MINIMUM χ^2
(ALSO TRUE IN HETEROSCEDASTIC CASE)

• • • Now, TO FIND MAXIMUM LIKELIHOOD

$$\frac{\partial}{\partial \mu} [\ln(L)]_{\mu=\mu_0} = 0$$

$$-\frac{1}{2\sigma^2} \sum -2(x_i - \mu_0) = 0$$

$$\frac{1}{\sigma^2} \sum (x_i - \mu_0) = 0$$

$$\sum x_i = N\mu_0$$

$$\mu_0 = \frac{1}{N} \sum_{i=1}^N x_i$$

$\Rightarrow \mu_0$ IS MEAN OF ALL MEASUREMENTS!

Properties of MLE - Consistent Convergence

- Asymptotically Normal

- Best Possible Estimate
Given Data at Hand.

Example: Exponential Distribution

$$f(x|M(\beta)) = \frac{1}{\beta} e^{-x/\beta}$$

For a set $\{x_i\}$ of measurements, what is the most likely value of β ?

$$L = \prod_{i=1}^N \frac{1}{\beta} e^{-x_i/\beta}$$

$$= \frac{1}{\beta^N} \prod_{i=1}^N e^{-x_i/\beta}$$

$$\{ \text{Now, } e^a e^b e^c \dots = e^{(a+b+c+\dots)} \}$$

$$= \frac{1}{\beta^N} e^{-\left(\sum_{i=1}^N x_i\right)/\beta}$$

$$l = \ln(L) = -N \ln \beta - \frac{1}{\beta} N \bar{x}$$

$$\left. \frac{dl}{d\beta} \right|_{\beta=\beta_0} = \frac{-N}{\beta_0} + \frac{N}{(\beta_0)^2} \bar{x} = 0$$

$$\boxed{\beta_0 = \bar{x}}$$

↳ Recall $L_{\text{true}} \neq 7_0$!